



735 samples - Detection of α -synuclein oligomers in CSF using the 24h Seed Amplification Assay (SAA).

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PPMI Project ID: 237

Summary

The α -synuclein (α Syn) Seed Amplification Assay (α S-SAA) takes advantage of the intrinsic self-replicating nature of α Syn misfolded aggregates (seeds) and multiplies them for their detection in synucleinopathy cases such as Parkinson's disease (PD), multiple system atrophy (MSA), and dementia with Lewy bodies (DLB). Amplification occurs *in vitro* when α Syn seeds in cerebrospinal fluid (CSF) are fragmented to generate numerous smaller seeds that grow by incorporating native recombinant protein. Given the nature of the technique, minute quantities of α Syn seeds in human bodily fluids can be multiplied and thereafter detected by conventional methods, such as immunoassays or fluorescence. Following this principle, α S-SAA was standardized for the highly sensitive and specific detection of α Syn seeds in CSF from patients with PD and other synucleinopathies¹⁻³. Amprion's α S-SAA has been further optimized to differentiate MSA from Lewy body disease (LBD) (PD and DLB) based on maximum fluorescence, use a decreased volume of recombinant α Syn, and the assay length has been shortened to 24h (24h α S-SAA) while maintaining similar sensitivity and specificity (~95%) of LBD and control samples, respectively.

Method

24h α S-SAA

The reaction mixture for amplification of endogenous α Syn seeds in CSF has a final volume of 100 μ L and consists of 100mM PIPES pH 6.5, 0.5M NaCl, 0.1% sarkosyl, 10 μ M ThT, 0.3mg/mL recombinant α Syn, and 40 μ L CSF. 2x 1/8" silicon nitride beads are included per reaction. The reaction is performed in 96-well plates and the samples are analyzed in 3 horizontal replicates (kinetic parameters are reported per replicate (Rep1, Rep2, Rep3)). The plate is incubated at 42°C and shaken for 1 min every 14 min. Fluorescence is read at 440nm-Ex/490nm-Em as relative fluorescence units (RFU) after every shaking cycle for 97 cycles (24h). Under these conditions, CSF samples containing α Syn seeds display robust aggregation curves within 24h, while CSF samples from control patients do not. This runtime is significantly shorter than both the modified assay described by Concha and collaborators⁴, which took 150h and was previously uploaded as "Amprion- α S-SAA" to the LONI database, and the original assay described by Shahnawaz and collaborators⁴, which took 300h and was previously uploaded as "Amprion slow α S-PMCA" to the LONI database. In this project, **735** deidentified CSF samples were analyzed over 1 week using a single lot of recombinant α Syn substrate and a high-throughput system consisting of 2 FLUOstar Omega shaker/readers (IDs 2, and 4) coupled to a robotic arm that shuffles 8 plates between a





storage magazine, where the plates incubate, and a designated FLUOstar Omega, where the plates are shaken and the fluorescence is read. The system is set up inside an incubator set to 42°C. Initial analysis of the deidentified samples was completed over the course of 3 runs of 8 plates. A fourth run of a single plate was necessary for reanalysis of a select group of samples. 4 PD and 4 HC controls were provided identified by PPMI and one set was analyzed per run, making the total number of analyzed samples 743.

Data analysis

Amprion-24h α S-SAA Qualitative.

α Syn amplification normally presents a classic amyloid aggregation curve in the 24h α S-SAA, including a lag phase, an exponential growth, and a final plateau. Samples without aggregation display background fluorescence values, but no aggregation curve. Amprion developed an algorithm to make a final qualitative determination of samples using the information from all 3 replicates: samples where all 3 replicates have a maximum fluorescence (F_{max}) $\geq 3,000$ RFU, or exactly 2 replicates have a $F_{max} \geq 3,000$ RFU and $< 45,000$ RFU, are called “Positive” and α Syn seeds are detected, while samples where exactly 0 or 1 replicate has a $F_{max} \geq 3,000$ RFU are called “Negative” and seeds are not detected. In borderline cases where only 2 replicates have a $F_{max} \geq 45,000$ RFU and 1 replicate has a $F_{max} < 3,000$ RFU, or 1 replicate has a $F_{max} \geq 45,000$ RFU and 1 replicate has a $F_{max} \geq 3,000$ RFU and $< 45,000$ RFU and 1 replicate has a $F_{max} < 3,000$ RFU, a definitive call cannot be made based on the three replicates and the sample is deemed “Inconclusive”, and reanalysis is recommended. Both inconclusive and conclusive retest results are reported. “Retest” in the Comments column indicates the retest results for a given sample. Since data is requested to be reported per replicate, qualitative results are included for all 3 replicates (QUALRep1, QUALRep2, QUALRep3) (Table I). Thus, qualitative results for Rep1-3 are the same since the algorithm uses the information from all 3 replicates to make a final determination.

Amprion-24h α S-SAA Synucleinopathy Status.

In addition to qualitative results, the 24h α S-SAA also generates Synucleinopathy Status results. Amprion's algorithm makes a final determination of samples using the information from all 3 replicates to differentiate “Positive” samples into “Type1” and “Type2” aggregation based on F_{max} . “Negative” samples have a Synucleinopathy Status of “N.A”, while “Inconclusive” samples have a Synucleinopathy Status of “Undetermined”. “Positive” samples where all 3 replicates have a $F_{max} \geq 45,000$ RFU are called “Type1”, while “Positive” samples where 2 or 3 replicates have a $F_{max} < 45,000$ RFU are called “Type2”. In rare cases where a “Positive” sample has exactly 2 replicates with a $F_{max} \geq 45,000$ RFU, a definitive call cannot be made based on the three replicates and the synucleinopathy status of the sample is deemed “Undetermined”, and reanalysis is recommended. Both inconclusive and conclusive retest results are reported. Since data is requested to be reported per replicate, synucleinopathy status results are included for all 3 replicates (SynucleinRep1, SynucleinRep2, SynucleinRep3) (Table I). Thus, synucleinopathy status results





for Rep1-3 are the same since the algorithm uses the information from all 3 replicates to make a final determination.

α S-SAA amplification parameters.

Fluorescence traces were further analyzed to estimate kinetic parameters that characterize the amplification curve of each replicate, such as the maximum fluorescence (Fmax, [RFU]), the maximum slope value obtained for a given assay reaction (Smax, [RFU/s]), time to Smax (TSmax, [hours]), the time to threshold (TTT, [hours]), and the area under the curve (AUC, [RFU*s]) (Table I).

Reported exceptions.

Some vials did not have enough volume and the last replicate was analyzed with less than 40 μ L of sample, which is reported as an exception for each replicate (SampleVolRep1, SampleVolRep2, SampleVolRep3) (Table I). Values for SampleVol: “NEV1” (Not Enough Volume 1) for CSF samples that contained between 20-30 μ L of the volume required for the third replicate; “NEV2” (Not Enough Volume 2) for CSF samples that contained more than 30 μ L but less than 40 μ L; and “NEV3” (Not Enough Volume 3) is for CSF samples that contained less than 20 μ L. In one case, the provided sample volume was very low, and only 2 replicates were able to be analyzed; the value for SampleVol for the third replicate is “NCS”, indicating no CSF sample.

Table I. Description of TESTNAMES and outcome for the data uploaded to LONI.

TESTNAME	Outcome
Amprion-24 α S-SAA Qualitative (QUALRepX ¹)	Character: Positive, Negative, Inconclusive. All replicates have the same value.
Amprion-24 α S-SAA Synucleinopathy Status (SynucleinRepX ¹)	Character: Type1, Type2, Undetermined, N.A. All replicates have the same value.
Amprion-24 α S-SAA Fmax (FmaxRepX ¹)	Numeric [RFU]. Each replicate has a different value.
Amprion-24 α S-SAA Smax (SmaxRepX ¹)	Numeric [RFU/s]. Each replicate has a different value.
Amprion-24 α S-SAA TSmax (TSmaxRepX ¹)	Numeric [hours]. Each replicate has a different value.
Amprion-24 α S-SAA TTT (TTTRepX ¹)	Numeric [hours]. Each replicate has a different value. Replicates without amplification do not have a TTT value (n.a.).
Sample volume (SampleVolRepX ¹)	Character: “NEV1” (20 to 30 μ L), NEV2 (30 μ L to <40 μ L), NEV3 (<20 μ L). Indicates a replicate did not have the 40 μ L specified per reaction.
Amprion-24 α S-SAA AUC (AUCRepX ¹)	Numeric [RFU*s]. Each replicate has a different value.
Amprion-24 α S-SAA Instrument (InstrumentRepX ¹)	Numeric. Instruments were identified as 2 and 4.

¹RepX represents Rep1, Rep2, and Rep3.

Results

735 deidentified samples along with 4 PD and HC identified controls were received and analyzed. Of the 735 deidentified CSF samples analyzed, there were 724 samples (97.7%) with a conclusive





result, 2 samples (0.3%) with Positive-Undetermined results, and 15 samples (2.0%) with inconclusive results. Of the Inconclusives, 1 sample did not have sufficient volume for retest (Inconclusives/not-retested), but the other 14 Inconclusive samples were retested. Of the 14 retests of the Inconclusives, 9 produced a conclusive result (64.3%), while 5 produced inconclusive results once again. Of the 2 retests of the Positive-Undetermined samples, 2 produced a conclusive result (100%). Including retest results as final, there were 419 (57.0%) Positives, 310 (42.2%) Negatives, 1 (0.1%) Inconclusives/not-retested, and 5 (0.7%) Inconclusives/retested. Of the Positives, 413 (98.6%) had Type1 aggregation and 6 (1.4%) had Type2 aggregation. All 4 PPMI-provided PD control pools and HC control pools produced Positive-Type1 and Negative results, respectively.

References

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3. Arnold, M. R. *et al.* α -Synuclein Seed Amplification in CSF and Brain from Patients with Different Brain Distributions of Pathological α -Synuclein in the Context of Co-Pathology and Non-LBD Diagnoses. *Ann. Neurol.* (2022) doi:10.1002/ana.26453.
4. Shahnawaz, M. *et al.* Development of a Biochemical Diagnosis of Parkinson Disease by Detection of α -Synuclein Misfolded Aggregates in Cerebrospinal Fluid. *JAMA Neurol.* **74**, 163 (2017).

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